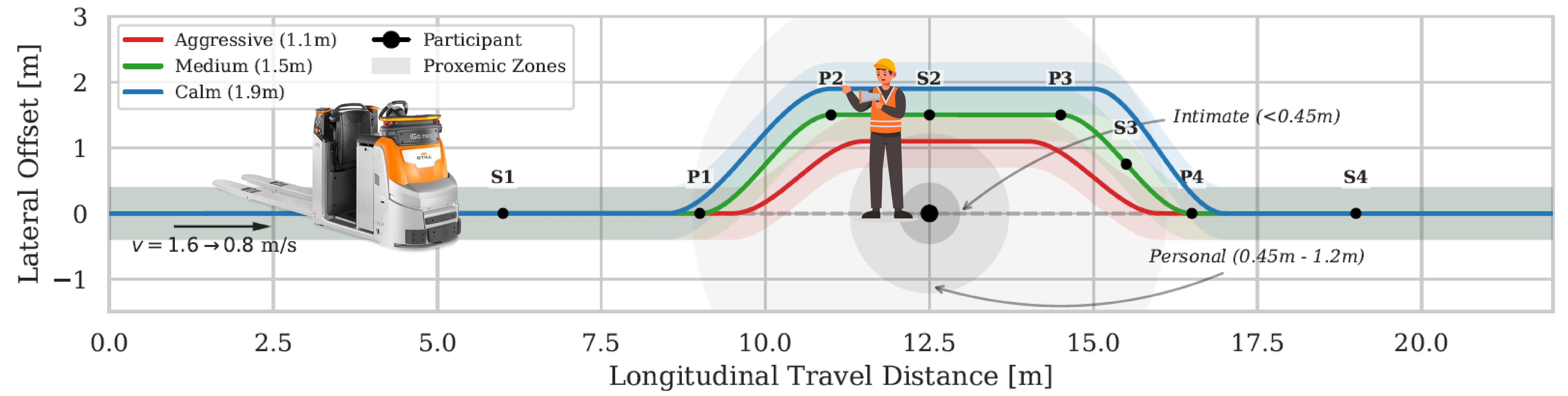


Perceived Safety of Workers in Encounters with Large Industrial AGVs

To what extent does a "Reality Gap" exist between simulated VR safety perceptions and actual real-world factory floor AGV interactions?



Experimental Design

Two Environments, One Encounter

Real World

Virtual Reality

AGV iGo neo OPX **Velocity** [0.8, 1.6] m/s

Stack ROS2 + RPP + eHMI **VR** HTC XR Elite + Unity

An AGV + eHMI interface (LED and Speaker) passes a stationary worker in 3 trajectory profiles. Encounter is replicated in VR with realistic sounds and AGV model

Methods

Measuring Perceived Safety

Continuous Trigger
Pressure sensitive VR controller samples perceived risk as $T(s) \in [0,1]$ during the encounter, causing unique spikes

Subjective Questionnaires
Post trial interviews. Each Questionnaire per trajectory profile. One final Questionnaire after all trials

Additional Metrics
Trigger rate (time engaged);
AUC = $\int T(s) ds$ (Cumulative threat over AGV path)

Profile	Modality	Trigger rate [%]	AUC	t_{total} [s]
Aggressive	Real	75	2.0 ± 2.0	3.8 ± 3.7
	VR	90	3.5 ± 3.3	6.4 ± 4.8
Medium	Real	50	0.7 ± 1.3	1.1 ± 2.3
	VR	40	0.6 ± 1.7	1.8 ± 3.4
Calm	Real	20	0.7 ± 2.0	0.9 ± 2.5
	VR	30	0.9 ± 2.0	2.4 ± 4.7

Combining real-time signal with post-hoc questionnaires captures moment of perceived threat and personal preference.. Allows for comparison across VR and Real-World

AGV Trajectory, Workers Preferences

Preferred lateral passing distance

1.5 – 2m

— VP-defined — Aggressive — Medium — Calm — Participant

Lateral Offsets
Match the **medium trajectory profile**

Longitudinal Onsets
Match the **calm trajectory profile**

Self parametrized paths converge on the lateral offset of the medium trajectory profile, with longitudinal onsets workers wanted space sideways and early signs of intent

